BATTERY PACK ON PCB: MODULAR 16S / 14S BREAKOUT & TRAINING BOARD
==========

- Form Factor: True A4 Landscape (297 mm x 210 mm)
- Power Stackup: 4-Layer Symmetrical Heavy Copper (2oz Top, 1oz Inner 1, 1oz Inner 2, 2oz Bottom)
- Safety Standard: IEC 62368-1 SELV (<=60V DC default lab rating with 14S; 16S under acrylic cover)
- Sensing: 4-Wire True Kelvin Voltage Sense Taps & 8x Dual-Can Saddle NTC Thermistors
- Active Safety: Zero-Drain Reverse Polarity Triads, 1500A High-Breaking Ceramic Fuse, 70V TVS Clamping

01_Cell_Module_Row1

File: 01_Cell_Module_Row1.kicad_sch

Module 1 (Cells 1 to 8): SMT 18650 Holders [C2988620], Banana Jacks, Reverse Triads

02_Cell_Module_Row2

File: 02_Cell_Module_Row2.kicad_sch

Module 2 (Cells 9 to 16): SMT 18650 Holders [C2988620], Banana Jacks, Reverse Triads, Inter-Row Bus

03_BMS_Interface_and_Thermal

File: 03_BMS_Interface_and_Thermal.kicad_sch

BMS Interface Bay: J102 17-Pin Voltage Harness + 17x PPTC [C207035] + 16x TVS [C266332]; J103 16-Pin + 8x NTCs [C123378]

04_Power_Path_and_Protection

File: 04_Power_Path_and_Protection.kicad_sch

Power Path Bay: F1 20A [C268205], Blown Fuse Alert, Pre-Charge Branch, 70V TVS [C1976987], >42V HV Strobe

05_Mechanical_and_Hardware

File: 05_Mechanical_and_Hardware.kicad_sch

Mechanical Bay: 6x Galvanically Isolated M3 Standoffs, Fiducials, High-Current Modular Tap Testpoints

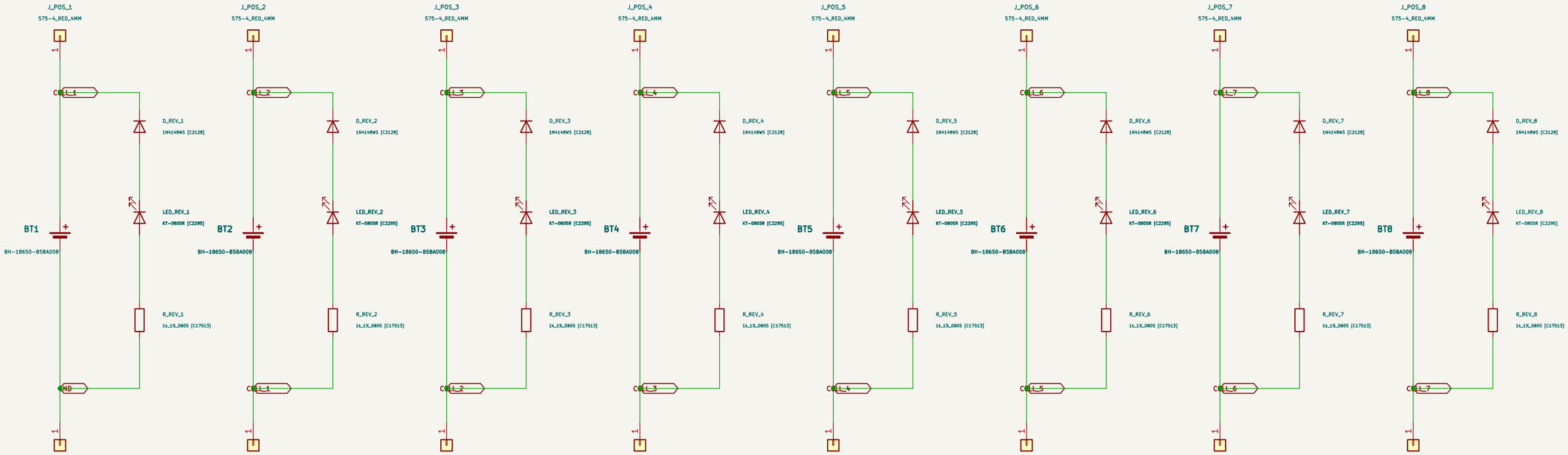
Sheet 1 of 6
Industry-Standard Modular 16S / 14S Battery Pack on PCB
IEC 62368-1 SELV <=60V | IEC 62619 | IEEE 1625 | IPC-2152
Antigravity Battery Engineering

Sheet: /
File: BatteryPack_16S_A4_Breakout.kicad_sch

Title: BATTERY PACK ON PCB — SYSTEM ARCHITECTURE & MASTER HIERARCHY

Size: A3	Date: 2026-10-12	**Rev: Rev 1.0**
KiCad E.D.A. 10.0.6		Id: 1/6

CELL MODULE 1: CELLS 1 TO 8 (UPPER CARRIER)
Symmetrical Layout • Across-Cell Banana Jacks • Zero-Drain Reverse Polarity Alert Triads

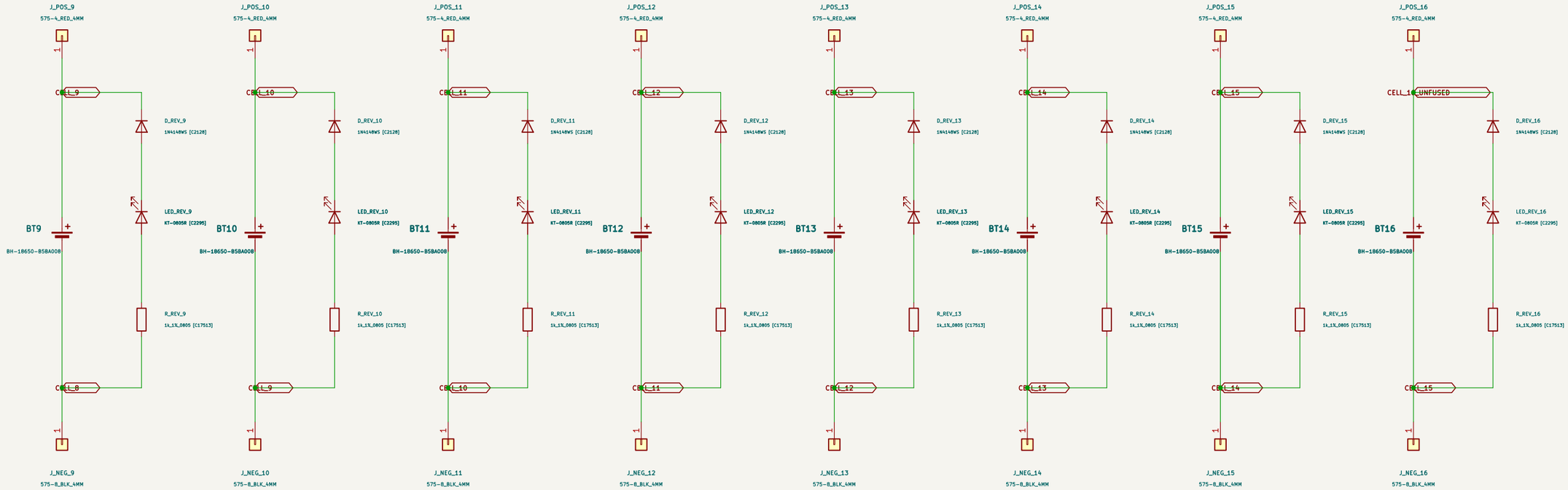


Sheet 2 of 6
Industry-Standard Modular 16S / 14S Battery Pack on PCB
IEC 62368-1 SELV <=60V | IEC 62619 | IEEE 1625 | IPC-2152
Antigravity Battery Engineering
Sheet: /01_Cell_Module_Row1/
File: 01_Cell_Module_Row1.kicad_sch

Title: CELL MODULE 1: CELLS 1 TO 8 (UPPER CARRIER)

Size: A3	Date: 2026-10-12	**Rev: Rev 1.0**
KiCad E.D.A. 10.0.6		Id: 2/6

CELL MODULE 2: CELLS 9 TO 16 (LOWER CARRIER)
Symmetrical Layout • Across-Cell Banana Jacks • Zero-Drain Reverse Polarity Alert Triads



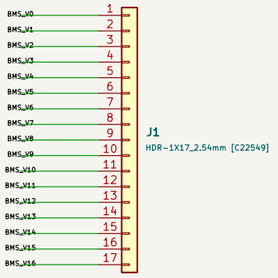
Sheet 3 of 6
Industry-Standard Modular 16S / 14S Battery Pack on PCB
IEC 62368-1 SELV <=60V | IEC 62619 | IEEE 1625 | IPC-2152
Antigravity Battery Engineering
Sheet: /02_Cell_Module_Row2/
File: 02_Cell_Module_Row2.kicad_sch

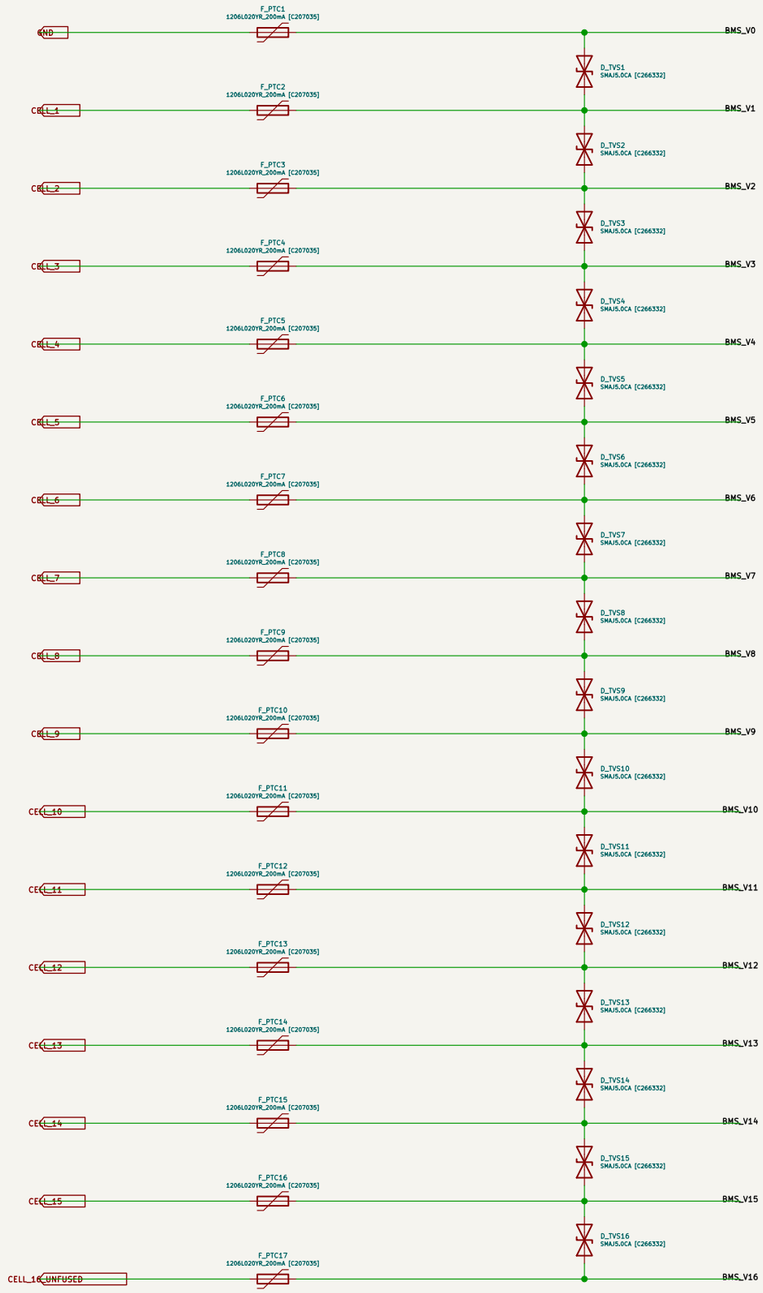
Title: CELL MODULE 2: CELLS 9 TO 16 (LOWER CARRIER)

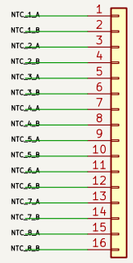
Size: A3	Date: 2026-10-12	**Rev: Rev 1.0**
KiCad E.D.A. 10.0.6		Id: 3/6

BMS INTERFACE BAY: VOLTAGE HARNESS (J102), PPTC FUSES & INTER-CELL DUAL-CAN NTC SENSING (J103)

17-Channel Resettable PTC Protection (200mA) [C207035] • 16x Differential Cell TVS Clamping [C266332] • 8x Dual-Can Saddle NTCs [C123378]

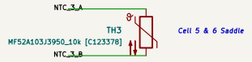




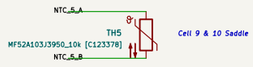


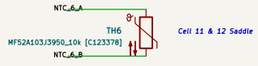


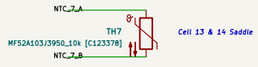














Sheet 4 of 6

Industry-Standard Modular 16S / 14S Battery Pack on PCB

IEC 62368-1 SELV <=60V | IEC 62619 | IEEE 1625 | IPC-2152

Antigravity Battery Engineering

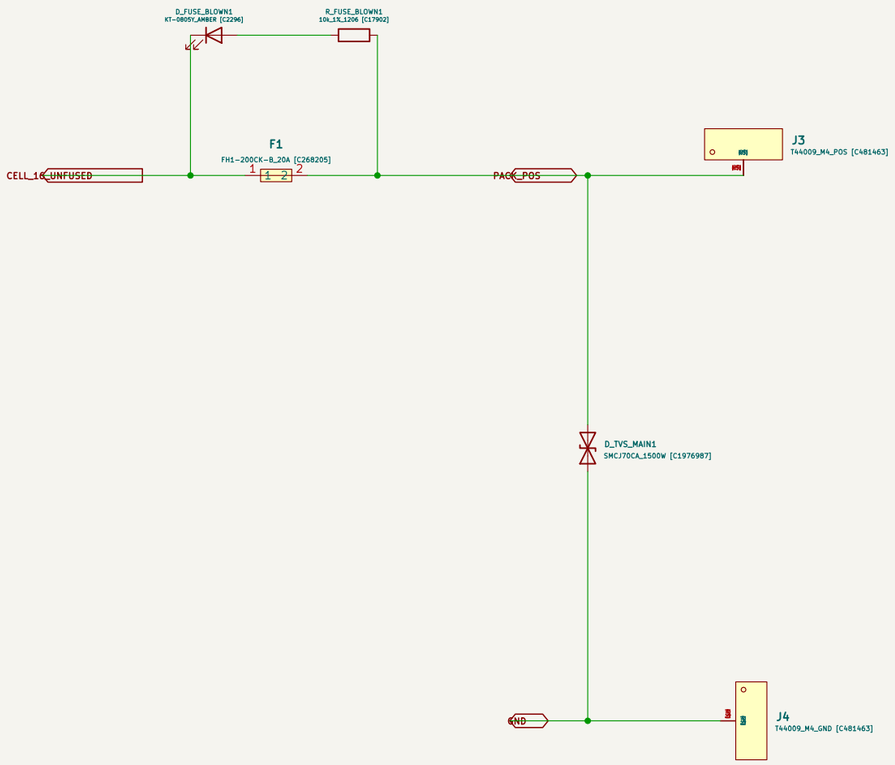
Sheet: /03_BMS_Interface_and_Thermal/

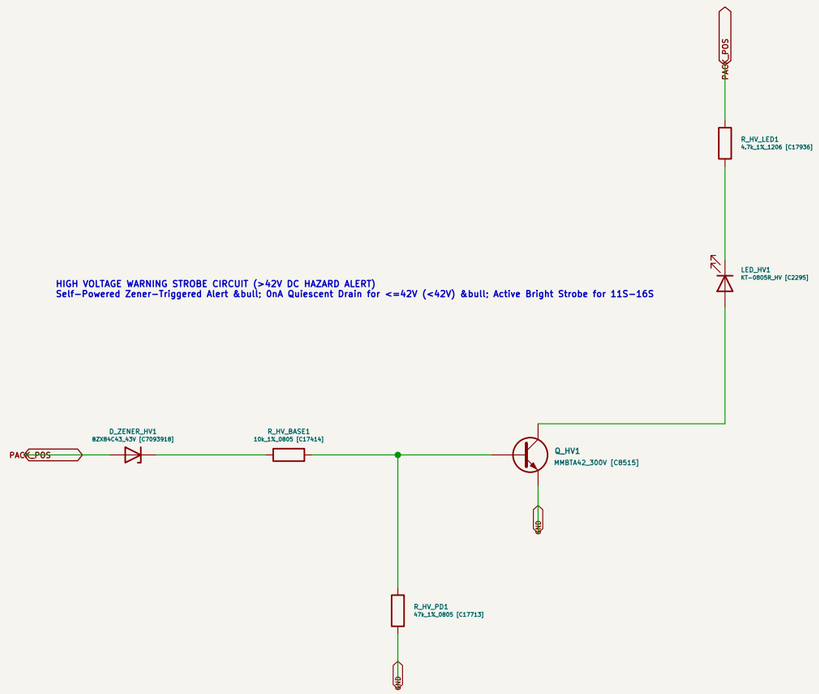
File: 03_BMS_Interface_and_Thermal.kicad_sch

Title: BMS INTERFACE BAY: VOLTAGE HARNESS, PPTCS & THERMAL MONITORING

Size: A3	Date: 2026-10-12	**Rev: Rev 1.0**
KiCad E.D.A. 10.0.6		Id: 4/6

POWER PATH, FUSING, IN-RUSH PRE-CHARGE & EDUCATIONAL STATUS BAY
Littelfuse 20A Ceramic Cartridge Fuse (1500A DC Breaking) [C268205] • Blown Fuse Alert LED • >42V HV Warning Strobe





Sheet 5 of 6
Industry-Standard Modular 16S / 14S Battery Pack on PCB
IEC 62368-1 SELV <=60V | IEC 62619 | IEEE 1625 | IPC-2152
Antigravity Battery Engineering
Sheet: /04_Power_Path_and_Protection/
File: 04_Power_Path_and_Protection.kicad_sch
Title: POWER PATH, FUSING, HV STROBE & EDUCATIONAL STATUS BAY

Size: A3	Date: 2026-10-12	**Rev: Rev 1.0**
KiCad E.D.A. 10.0.6		Id: 5/6

MECHANICAL HARDWARE BAY: ISOLATED STANDOFFS, FIDUCIALS & MODULAR TAP TESTPOINTS
6x Galvanically Isolated M3 Holes (Floating Ground) • Optical Fiducials • Keystone Color-Coded Test Points [C5199900 / C238122]

H1
M3_ISOLATED_HOLE

H4
M3_ISOLATED_HOLE

FID1
Fiducial_1mm

FID2
Fiducial_1mm

GND
TP_GND1
Keystone_5001_BLK [C238122]

CELL_7
TP_7S1
Keystone_5000_RED [C5199900]

CELL_10
TP_10S1
Keystone_5000_RED [C5199900]

H2
M3_ISOLATED_HOLE

H5
M3_ISOLATED_HOLE

CELL_14
TP_14S1
Keystone_5000_RED [C5199900]

PACK_POS
TP_16S1
Keystone_5000_RED [C5199900]

H3
M3_ISOLATED_HOLE

H6
M3_ISOLATED_HOLE

FID3
Fiducial_1mm

Sheet 6 of 6
Industry-Standard Modular 16S / 14S Battery Pack on PCB
IEC 62368-1 SELV <=60V | IEC 62619 | IEEE 1625 | IPC-2152
Antigravity Battery Engineering
Sheet: /05_Mechanical_and_Hardware/
File: 05_Mechanical_and_Hardware.kicad_sch
Title: MECHANICAL HARDWARE: ISOLATED STANDOFFS, TEST POINTS & FIDUCIAL

Size: A3	Date: 2026-10-12	**Rev: Rev 1.0**
KiCad E.D.A. 10.0.6		Id: 6/6